

Problem Statements

2.1 Background

Indian Railways operates one of the world’s most complex rail networks. Despite its scale, operational inefficiencies persist—certain stations are severely over-trafficked, many routes are chronically underserved, and scheduling patterns are sub-optimal. Without analytical visibility, infrastructure investments and policy decisions are made reactively rather than proactively.

2.2 Problem Statements Addressed

PS-1: Network Hub Identification

Which stations dominate train origination and termination, and how is network connectivity distributed?

PS-2: Route Efficiency Analysis

Which routes are fastest/slowest relative to their distance? Do longer routes maintain proportionally higher speeds?

PS-3: Service Frequency & Coverage Gaps

Which origin-destination pairs have fewer than 3 trains per week? Where are the scheduling blind spots?

PS-4: Class Availability & Passenger Equity

Are premium classes (1A, 2A) concentrated on certain corridors? Which classes are underrepresented?

PS-5: Intermediate Stop Complexity

Do trains with more stops also cover greater distances? How does stop count correlate with journey time and distance?

PS-6: Peak Departure Time Clustering

When do most trains depart? Are there overloaded scheduling windows causing operational stress?

2.3 Key Stakeholders will be

- Indian Railways Ministry & Policy Teams (scheduling and investment decisions)
- Station Operations Managers (platform and crew allocation)
- Passenger Advisory Bodies (equity and accessibility)
- Revenue Management Teams (class mix and pricing strategy)
- Data Engineering Teams (data quality and completeness)

2.4 Expected Outcomes

- A ranked list of strategic railway hubs requiring capacity investment
- Identification of underserved routes needing additional services
- Schedule rebalancing recommendations to reduce peak-hour congestion
- Passenger class equity analysis to guide coach allocation policy
- A production-ready Power BI dashboard for executive decision-making